PNOZmulti TCP/IP communication with SIMATIC PLCs in Step7 V5.5

Product
Type: PNOZmulti with Ethernet-Interface (TCP/IP)
Name: PNOZmulti series (complete list inside, please see Ch. 3.2, page 8)
Manufacturer: Pilz GmbH & Co. KG, Safe Automation

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Document Revision History

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01	2012-08-23	Creation	all
02	2014-06-16	Insert PNOZ m ES ETH Module	all

Validity of Application Note

This present Application Note is valid until a new version of the document is published.
This and other Application Notes can be downloaded in the latest version and for free from www.pilz.com.
For a simple search, use our content document (1002400) or the direct search function in the download area.

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June 2014

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1. Useful documentation

Reading the documentation listed below is necessary for understanding this application note. The availability of the indicated tools and safe handling are also presupposed with the user.

1.1. Documentation from Pilz GmbH & Co. KG

No.	Description	Item No.
1	Pilz international homepage, download section	www.pilz.com
2	PNOZmulti Communication Interfaces	1001154-EN-XX
3	PNOZmulti 2 Communication Interfaces	1002971-EN-XX
4	Application Note PNOZm Serial-2-Ethernet	1001958-EN-XX
5	Technical Catalogue PNOZmulti	1001153-EN-XX
6	Operating Manual PNOZ mm0.1p	1002101-EN-XX
7	Operating Manual PNOZ mm0.2p	1002235-EN-XX
8	Operating Manual PNOZ mmc1p	1002102-EN-XX
9	Operating Manual PNOZ m1p (ETH)	20878-EN-XX
10	Operating Manual PNOZ m B0	1002660-EN-XX
11	Operating Manual PNOZ m ES ETH	1002700-EN-XX

1.2. Documentation from other sources of information

No.	Description	Item No.
1	SIMATIC portal “Automation Systems” (international) Internet-Link to "www.automation.siemens.com"	
2	Document, Code and Step7-Project to topic “Wizard for creating connection data for open TCP/IP communication (Open Communication Wizard).” Internet-Link to "support.automation.siemens.com"	OC Wizard V2.3.3 Entrys-ID: 25209116

2. Preface

This document provides a Step by Step example how to communicate between Siemens S7-300 and PNOZ multi with TCP/IP connection.

Notice:
This Application Note describes a PNOZ m1p ETH, the PNOZ mmc1p is working in the same way. With a PNOZ m B0 ETH you have to take different Blocks (because of bigger Data Size) in the Simatic Example, it is also included in the Simatic Example.

The document is not intended as a technical documentation for general use of SIMATIC Manager Step 7. It may not be distributed to customers without a special note to its scope.

Notice: You need for this application these four projects

- PNOZmulti Configurator – provided Project
  - "**PNOZmulti_S7-PLC_ETH-TCP-IP_v2**"
  - "**PNOZ_mmc1p_S7-PLC_ETH-TCP-IP_v1**"
  - "**PNOZ_m_ES_ETH_S7-PLC_ETH-TCP-IP_v1**"
- Step7 – provided Project
  - "**PNOZm_S7-PLC_ETH_v2**"

This following Step7-Project is not provided. It is an example for your second target project.

- Step7 – Target Project Example "ETH_Multi_S7_ProfiNET_v2"

In addition to this manual the first **four projects**, which are described in the following, can be opened with a unzip program from the attached file

"**AN_PNOZm_ETH_TCP-IP_with_S7_1002671_SW_02.zip**"

by a double-click on this "Pin-Icon": » «

Notice:
It is not possible to open or save a *.zip File from the attachments with Adobe Reader.
>> For open or save this file, please use PDF-XChange Viewer (www.tracker-software.com).

Alternatively the file can be obtained from Pilz Technical Support (mailto:support@pilz.de)
Subject: Application Note 1002671

3. Configuration

3.1. List of used hardware

This Application Note has been tested with the following hardware:

- Siemens Simatic PLC: CPU 318-3FL01-0AB0 V3.2
- Pilz Configurable Control System: PNOZ m1p ETH 773104 Version v2.1
- Pilz Configurable Control System: PNOZ m B0 772100 Version v1.1
- Pilz Configurable Control System: PNOZ m ES ETH 772103 Version v1.0

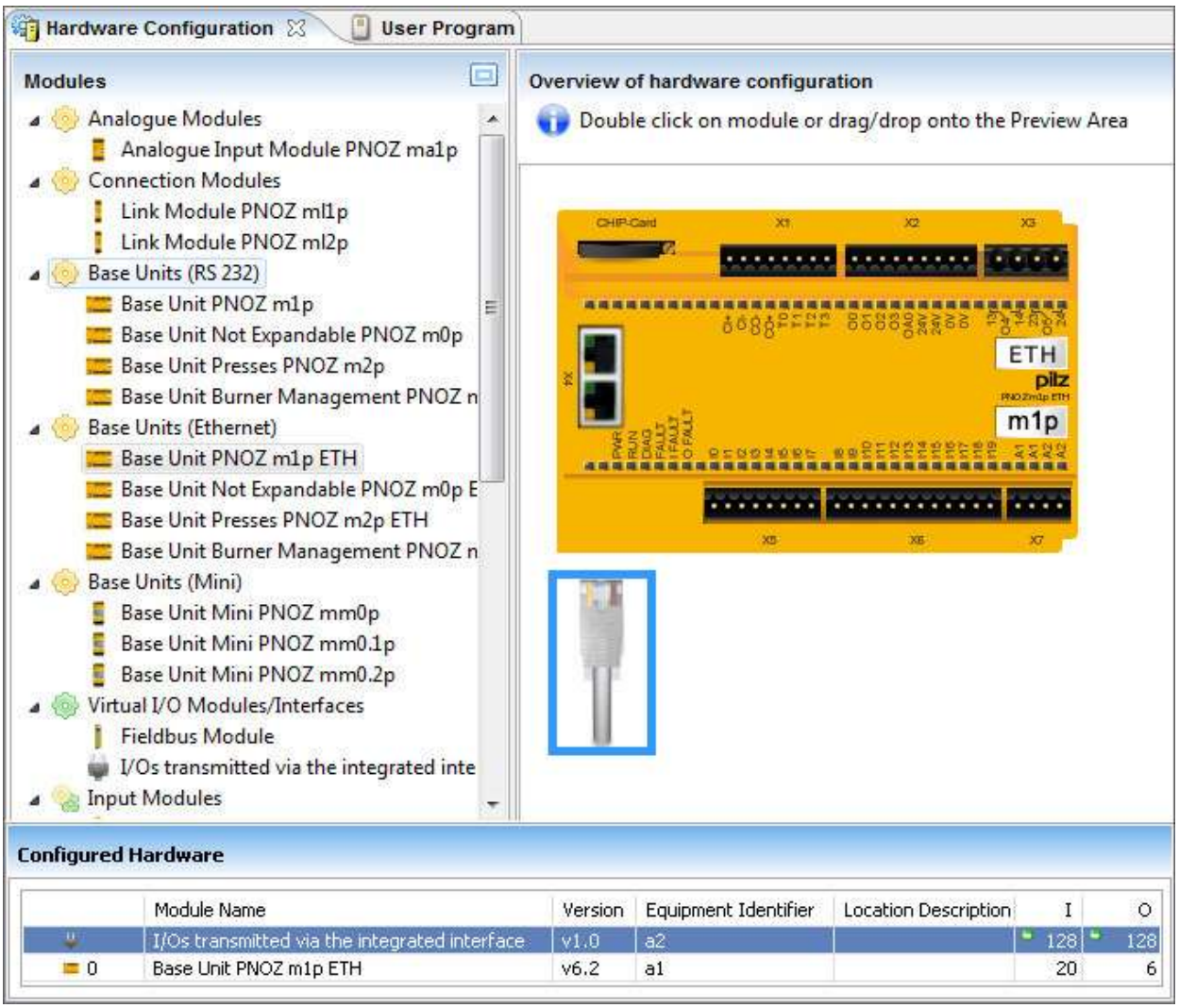


Fig. 1: PNOZmulti Configurator - Hardware Configuration PNOZ m1p ETH

3.2. List of supported Pilz hardware

- PNOZmulti Mini Series
  - PNOZ mm0.1p 772001 with PNOZ mmc1p ETH 772030
  - PNOZ mm0.2p 772002 with PNOZ mmc1p ETH 772030

- PNOZmulti 1st Series with integrated Ethernet ports
  - PNOZ m1p ETH 773103
  - PNOZ m3p ETH 773126
  - PNOZ m0p ETH 773113
  - PNOZ m2p ETH 773123
  - PNOZ m1p ETH coated version 773104

- PNOZmulti 1st Series with external Ethernet ports

Notice: The description of different structure of data telegram of the following configurations is not part of this document. For more information please see Ch. 1.1-No. 2, page 5.

  - PNOZ m1p base unit 773100
  - PNOZ m1p base unit coated version 773105
  - PNOZ m2p base unit press function 773120
  - PNOZ m3p base unit burner function 773125

- PNOZmulti 2nd Series B0 with (external) Ethernet ports
  - PNOZ m B0 772100
  - PNOZ m ES ETH 772130

3.3. List of used Software

This Application Note has been tested with the following software:

- Pilz PNOZmulti Configurator, Version 9.2.0
- Siemens Simatic STEP7, Version 5.5 SP3
- Siemens Simatic Open Communication Wizard, Version 2.3

3.4. List of used IP addresses

The following IP addresses has been used in the example project:

- Computer: 169.254.060.099
- PNOZmulti ETH: 169.254.060.001
- PNOZ m ES ETH: 169.254.060.002
- Simatic S7 PLC: 169.254.060.151

Notice: For more information about setting IP address open in software “PNOZmulti Configurator” Help function "Contents and Index" and choose "Create or edit project".

3.5. List of used S7 and PNOZ multi Programs according to PNOZ multi Hardware

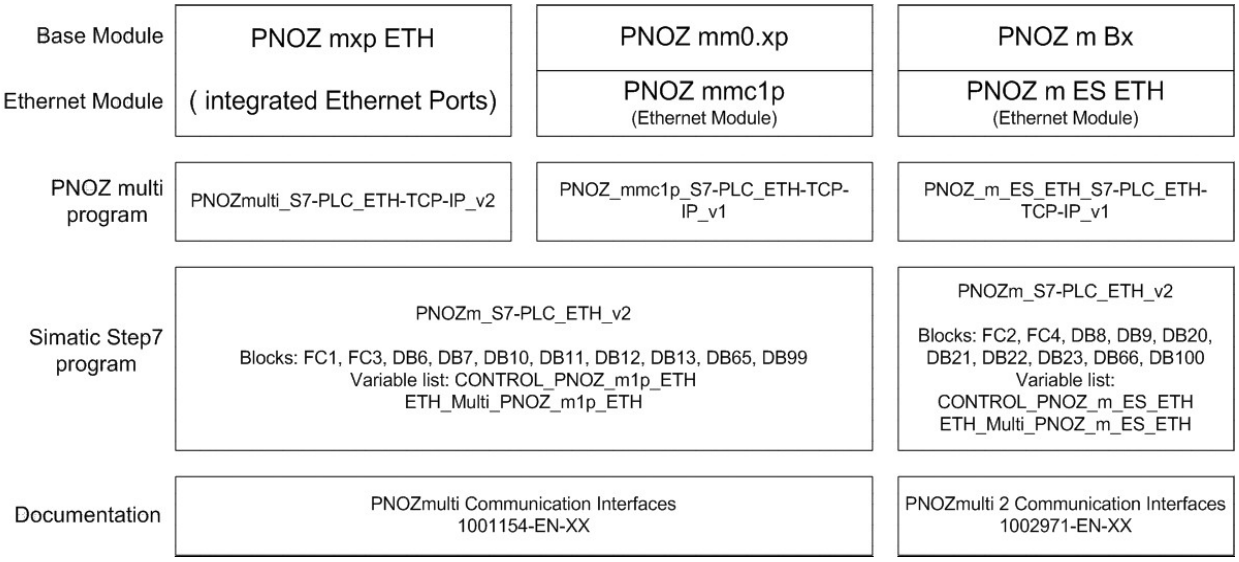


Fig. 2: List of used Programs

4. Application Task

Create a connection with TCP/IP and control communication between PNOZ m1p ETH / PNOZ m ES ETH and Siemens S7-300 PLC.
The initial S7-Hardware configuration is not part of this “Application Note”. One can get further details in the manuals about Siemens Simatic Step 7 (please see Ch. 1.2, page 5).

4.1. Setting IP address on PNOZmulti base unit

- Start the PNOZmulti Configurator and open “Scan Network”.



Fig. 3: PNOZmulti Configurator - Scan Network

- Choose your device and go online. For that click on the “Online”-button.

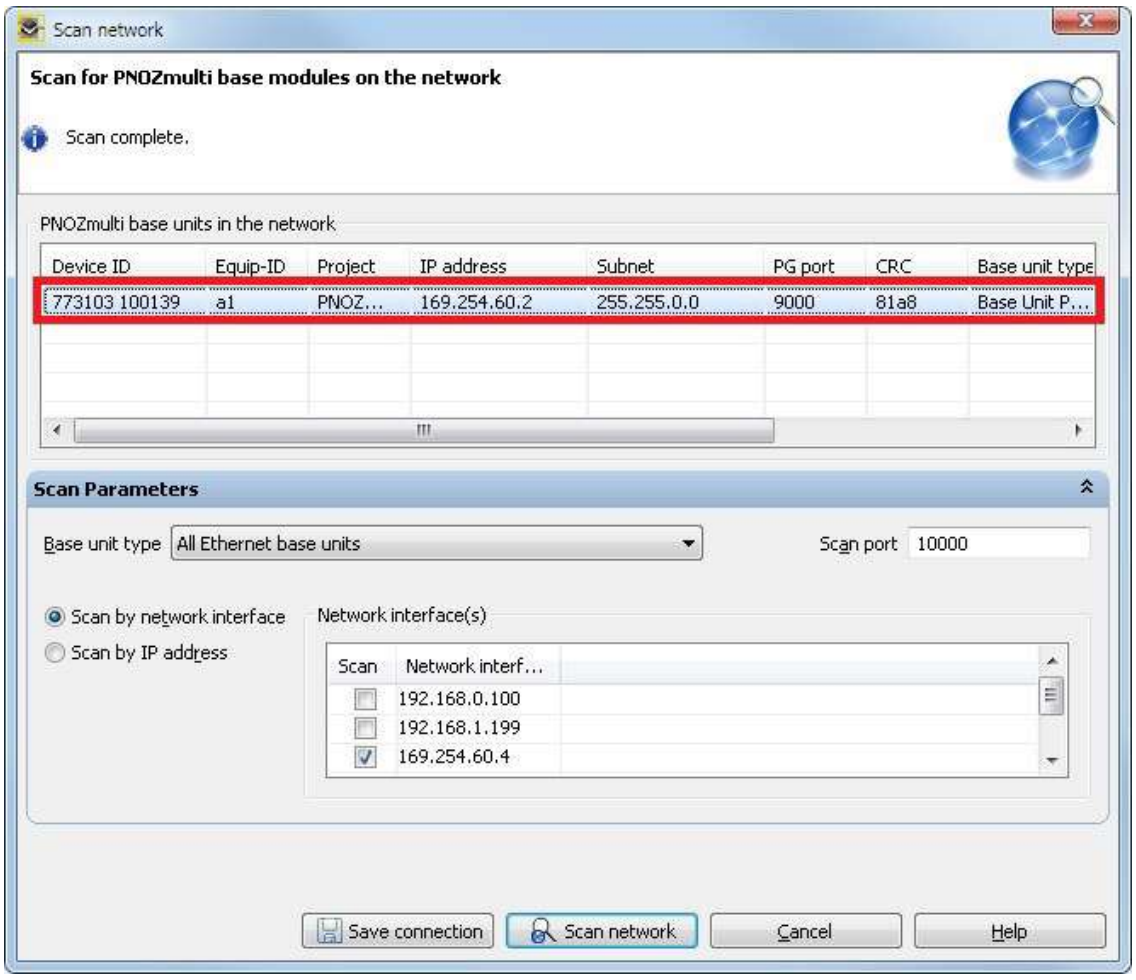


Fig. 4: PNOZmulti Configurator - Choose of the used device



Fig. 5: PNOZmulti Configurator - Go online with the device

If necessary adjust the IP address of your PNOZmulti.

- Continue with function "Configure Device Ethernet Connection".

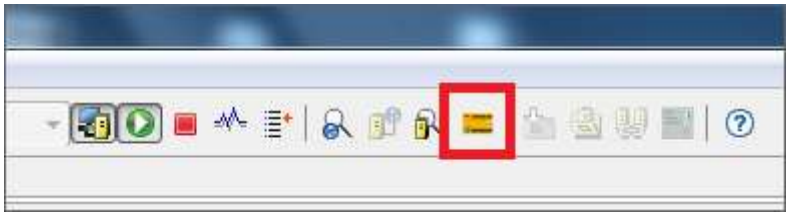


Fig. 6: PNOZmulti Configurator - Configure device Ethernet connection

- Change the IP address of your device und confirm with "OK".

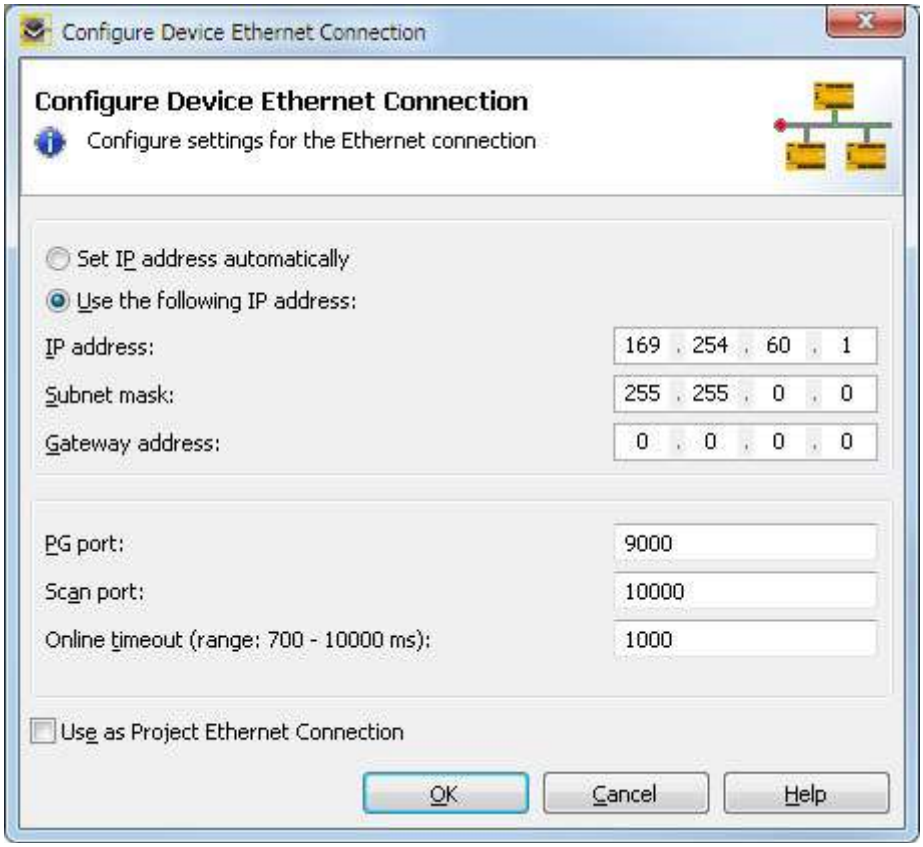


Fig. 7: PNOZmulti Configurator - Adapt the IP address on base unit

The defined IP address is now set to PNOZmulti. The device wills response to that IP address.

4.2. Setting IP address on Simatic PLC (ProfiNET)

- Open the provided Step7-Project “PNOZmulti_S7-PLC_ETH_v2” in the Simatic Manager.
- Copy the blocks this 1st project to your actual project (here as example “ETH_Multi_S7_ProfiNET_v2”) and click twice at “Hardware”.
- Click twice on PN-IO (X2). The window "Properties - PN-IO" open.

If necessary improve the device name to "PN-IO" **[1]**.

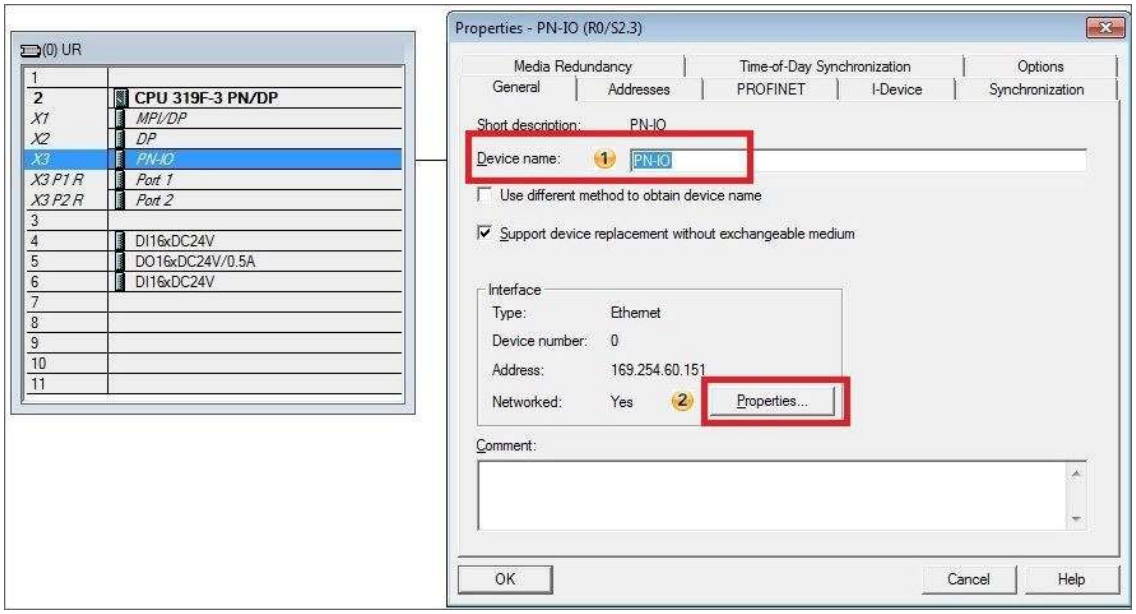


Fig. 8: Simatic HW Config - Setting device name

- Click on the Button "Properties…" if you have to change the IP address **[2]**.

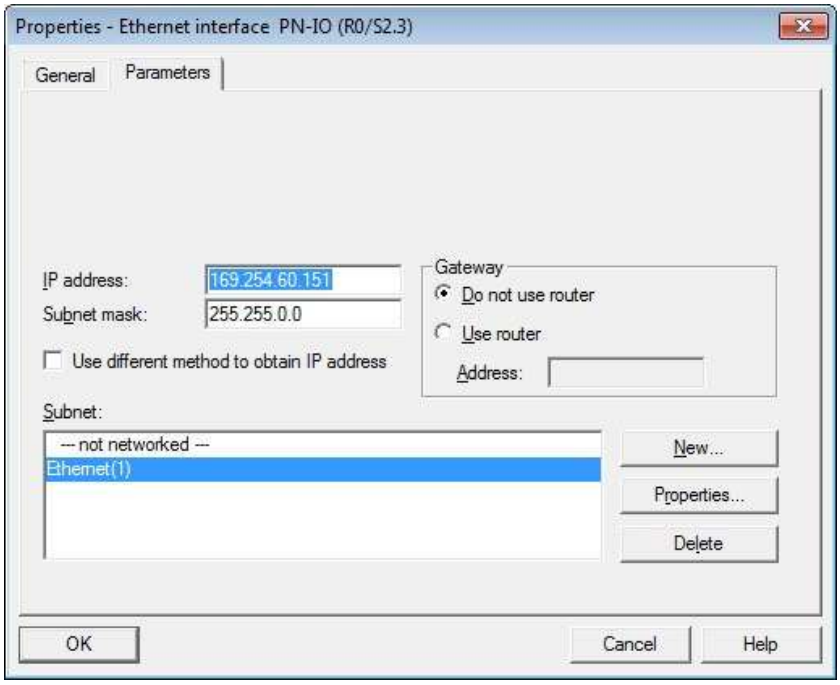


Fig. 9: Simatic HW Config - Change the IP-address

4.3. Settings with Open Communication Wizard

- Start the "Open Communication Wizard" and click at "Next".

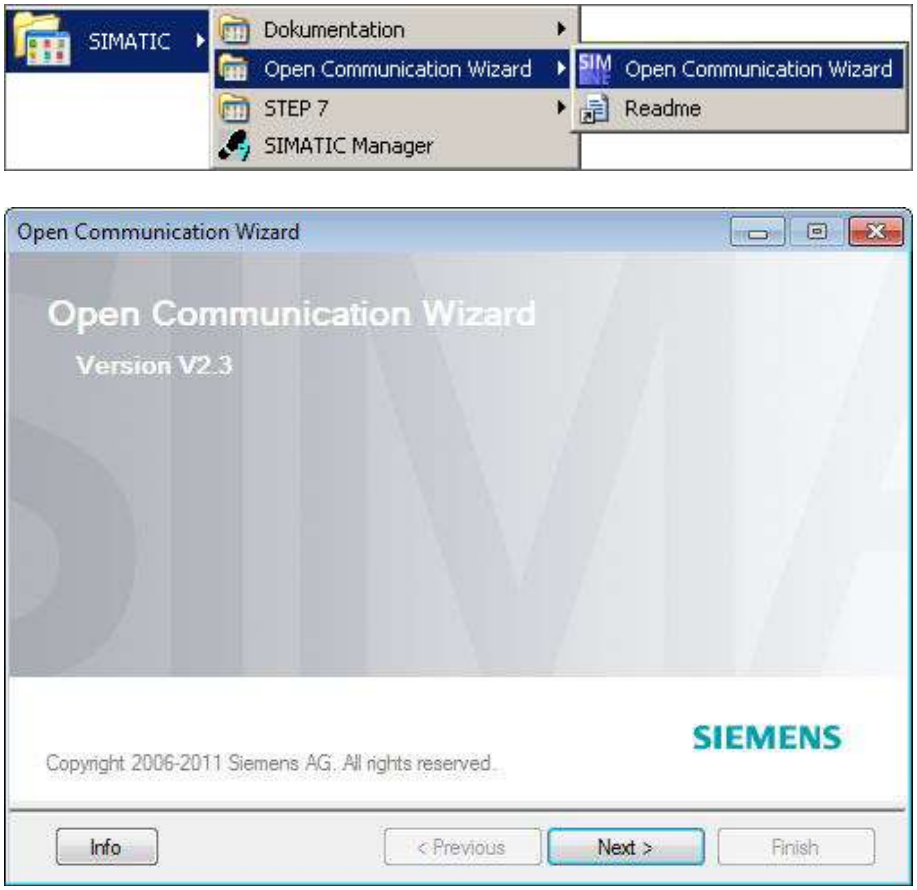


Fig. 10: OC Wizard - Start Open Communication Wizard

- Choose your S7 project and your block folder and confirm with "Open".
- Click "Next" to continue.

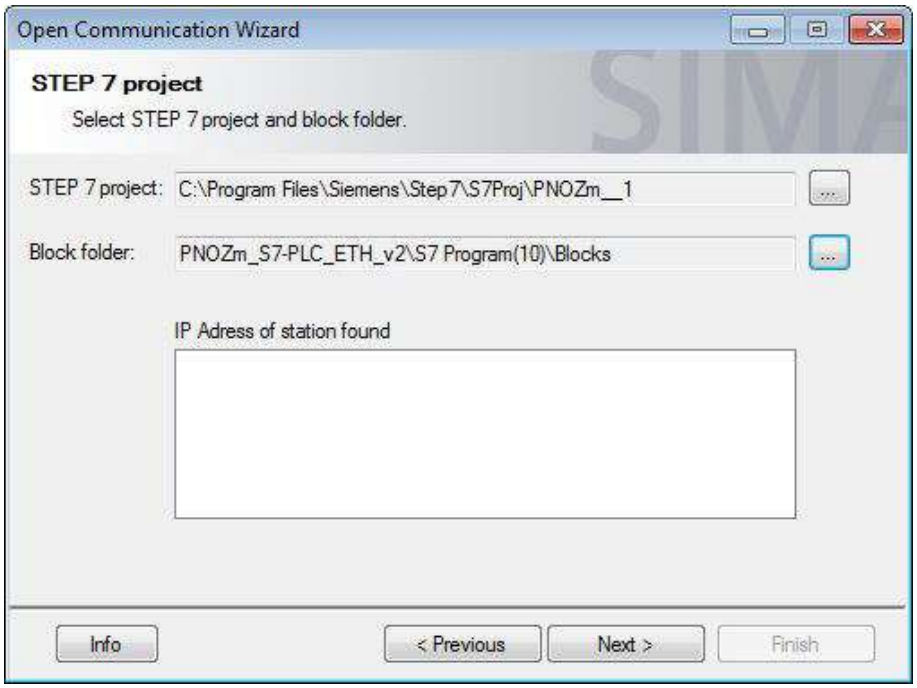


Fig. 11: OC Wizard - Choose the Step 7 project and block folder

- Click "Next" to continue.

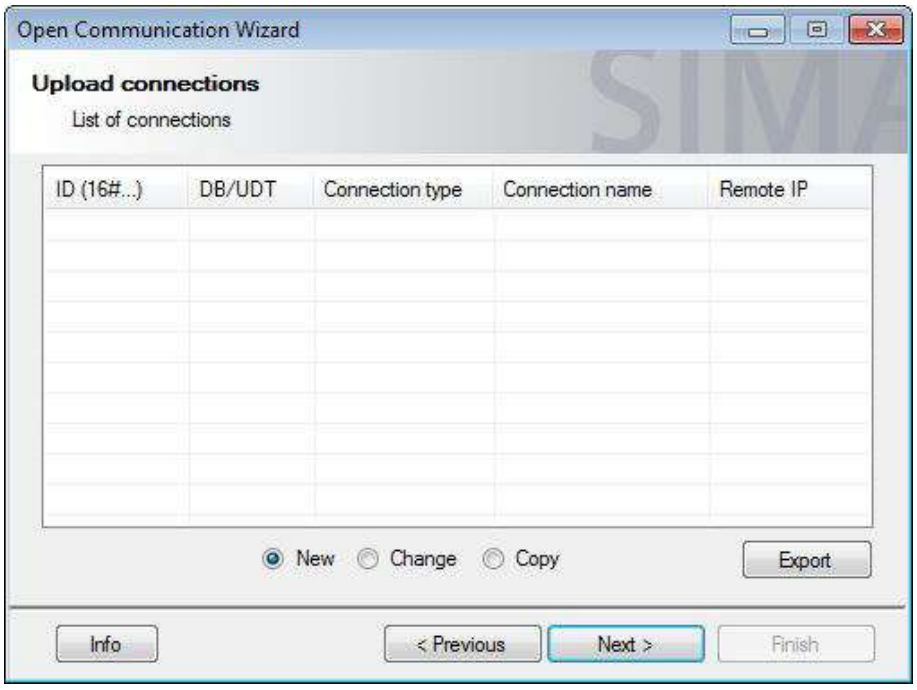


Fig. 12: OC Wizard – Check connection

- Choose "TCP native" and go on with "Next".

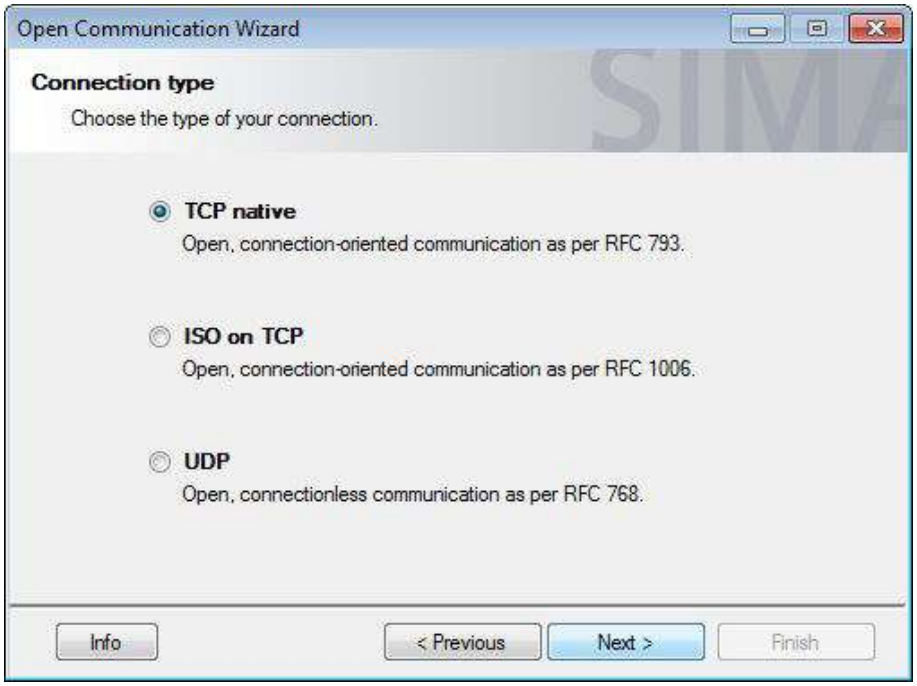


Fig. 13: OC Wizard – Choose connection type “TCP native”

- The “Communication partner B (here: PNOZmulti base unit) is not a S7-CPU”, so activate the right check box and continue with "Next".

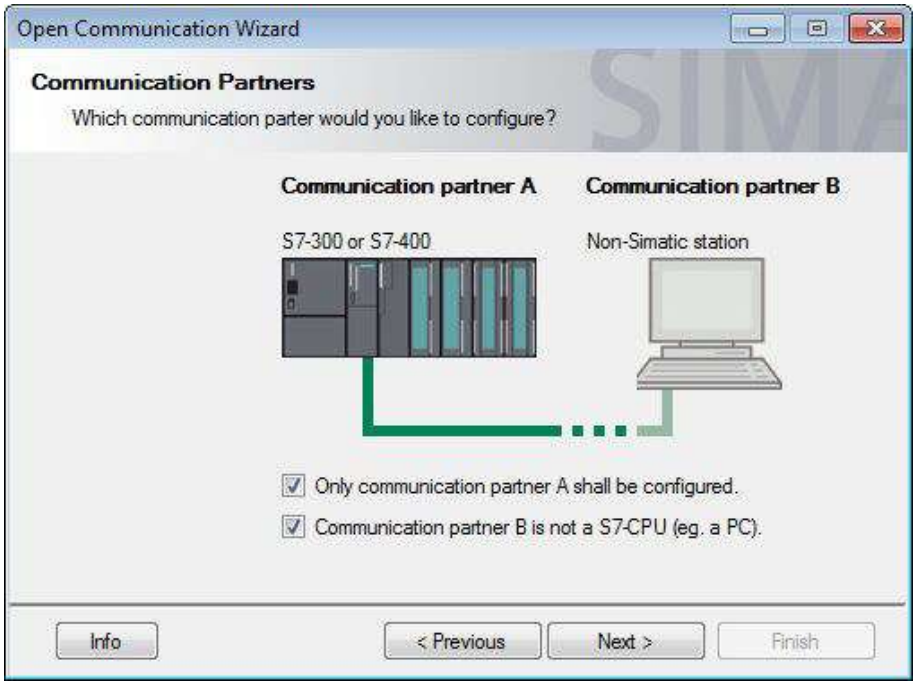


Fig. 14: OC Wizard – Set properties of “Communication partner B”